

$$\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 5x} = \lim_{x \rightarrow 0} \frac{\sin 4x}{\frac{\sin 5x}{\cos 5x}} = \lim_{x \rightarrow 0} \frac{\sin 4x \cos 5x}{\sin 5x} =$$

$$\frac{\sin(4 \cdot 0)}{\frac{\sin(5 \cdot 0)}{\cos(5 \cdot 0)}} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \rightarrow 1$$

$$\left(\cdot \frac{\frac{1}{x}}{\frac{1}{x}} \right) \lim_{x \rightarrow 0} \frac{\frac{\sin 4x}{x} \cos 5x}{\frac{\sin 5x}{x}} =$$

$$= \lim_{x \rightarrow 0} \frac{4 \cdot \frac{\sin 4x}{4x} \cos 5x}{5 \cdot \frac{\sin 5x}{5x}} \rightarrow \frac{4}{5}$$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \left(\frac{x^3 + 4x}{x^3 + 2x - 3} \right)^{x^2 - 3} &= \lim_{x \rightarrow \infty} \left(\frac{x^3 + 2x - 3 + 2x + 3}{x^3 + 2x - 3} \right)^{x^2 - 3} \\
 &= \lim_{x \rightarrow \infty} \left(1 + \frac{2x + 3}{x^3 + 2x - 3} \right)^{x^2 - 3} = \\
 &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\frac{x^3 + 2x - 3}{2x + 3}} \right)^{x^2 - 3} = \\
 &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{\frac{x^3 + 2x - 3}{2x + 3}} \right)^{\frac{x^3 + 2x - 3}{2x + 3}} \right)^{\frac{2x + 3}{x^3 + 2x - 3} (x^2 - 3)} \\
 &= e^{\lim_{x \rightarrow \infty} \frac{(2x + 3)(x^2 - 3)}{x^3 + 2x - 3}} \rightarrow e^2
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{(2x + 3)(x^2 - 3)}{x^3 + 2x - 3} &= \lim_{x \rightarrow \infty} \frac{2x^3 - 6x + 3x^2 - 9}{x^3 + 2x - 3} = \\
 \left(\cdot \frac{\frac{1}{x^3}}{\frac{1}{x^3}} \right) &= \lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x} - \frac{6}{x^2} - \frac{9}{x^3}}{1 + \frac{2}{x^2} - \frac{3}{x^3}} \rightarrow 2
 \end{aligned}$$

$$\int_1^2 x^3 \sqrt{2x^2 - 1} dx =$$

$$t = 2x^2 - 1 \Rightarrow x = \sqrt{\frac{t+1}{2}}$$

$$dt = d(2x^2 - 1) = (2x^2 - 1)' dx = 4x dx$$

$$x dx = \frac{dt}{4}$$

$$\begin{aligned} \int \left(\sqrt{\frac{t+1}{2}}\right)^2 \sqrt{t} \frac{dt}{4} &= \frac{1}{4} \int \frac{(t+1)\sqrt{t}}{2} dt = \frac{1}{8} \int t\sqrt{t} + \sqrt{t} dt = \\ &= \frac{1}{8} \left(\int t \cdot t^{\frac{1}{2}} dt + \int t^{\frac{1}{2}} dt \right) = \frac{1}{8} \left(\int t^{\frac{3}{2}} dt + \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right) = \\ &= \frac{1}{8} \left(\frac{t^{\frac{3}{2}+1}}{\frac{3}{2}+1} + \frac{t^{\frac{3}{2}}}{\frac{3}{2}} \right) = \frac{1}{8} \left(\frac{t^{\frac{5}{2}}}{\frac{5}{2}} + \frac{t^{\frac{3}{2}}}{\frac{3}{2}} \right) = \frac{1}{8} \left(2 \frac{t^{\frac{5}{2}}}{5} + 2 \frac{t^{\frac{3}{2}}}{3} \right) = \\ &= \frac{t^{\frac{5}{2}}}{20} + \frac{t^{\frac{3}{2}}}{12} = \frac{(2x^2-1)^{\frac{5}{2}}}{20} + \frac{(2x^2-1)^{\frac{3}{2}}}{12} + C = \\ &= \frac{(2x^2-1)^{\frac{3}{2}} (2x^2-1)}{20} + \frac{(2x^2-1)^{\frac{3}{2}}}{12} + C = \\ &= (2x^2-1)^{\frac{3}{2}} \left(\frac{2x^2-1}{20} + \frac{1}{12} \right) + C = \\ &= (2x^2-1)^{\frac{3}{2}} \left(\frac{3(2x^2-1)+5}{60} \right) + C = \\ &= (2x^2-1)^{\frac{3}{2}} \left(\frac{6x^2-3+5}{60} \right) + C = \\ &= \frac{(2x^2-1)^{\frac{3}{2}} (3x^2+1)}{30} + C \end{aligned}$$

$$\begin{aligned} &= \frac{(2x^2-1)^{\frac{3}{2}} (3x^2+1)}{30} \Big|_1^2 = \frac{(2 \cdot 2^2 - 1)^{\frac{3}{2}} (3 \cdot 2^2 + 1)}{30} - \\ &- \frac{(2 \cdot 1^2 - 1)^{\frac{3}{2}} (3 \cdot 1^2 + 1)}{30} = \frac{13\sqrt{7^3} - 4}{30} \end{aligned}$$